

Customer Churn Intelligence

Overview of churn patterns, customer risk and model performance

Churn Overview

Churn Drivers

Customer Risk

Total Customers

7,043

Actual Churn Rate

26,5 %

High-Risk Customers

3,235

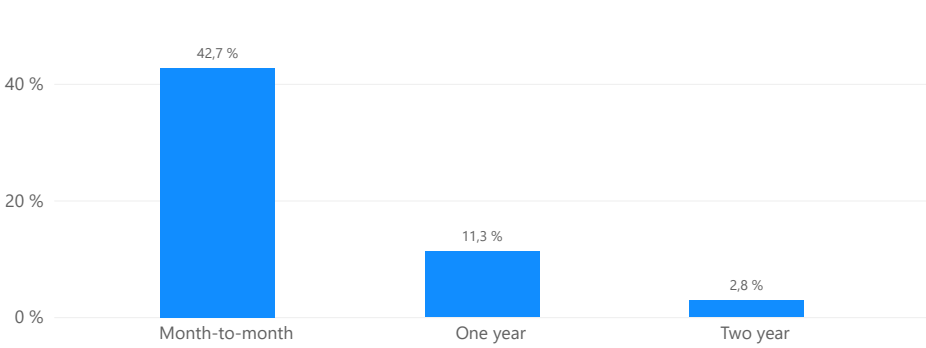
Average Churn Probability

39,3 %

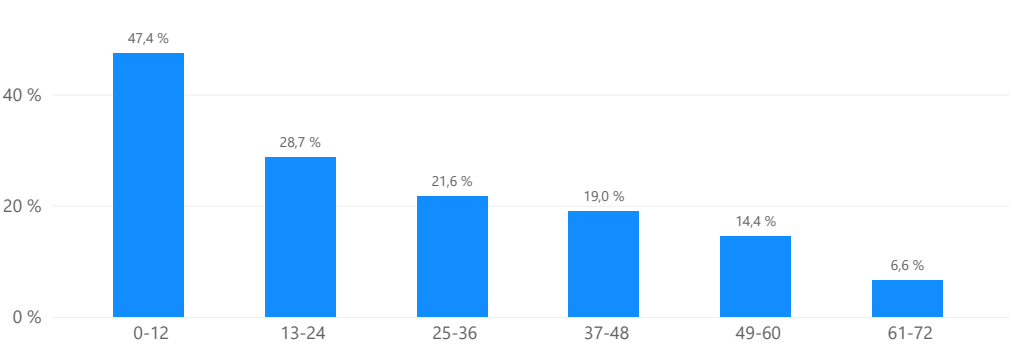
Model performance — held-out test set

Accuracy: 76.7% | Recall: 71.9% | F1: 62.1% | ROC-AUC: 84.2%

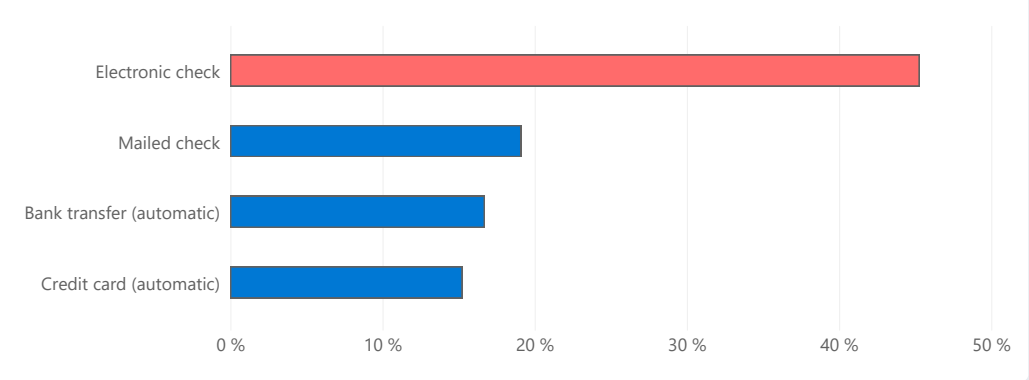
Churn Rate by Contract Type



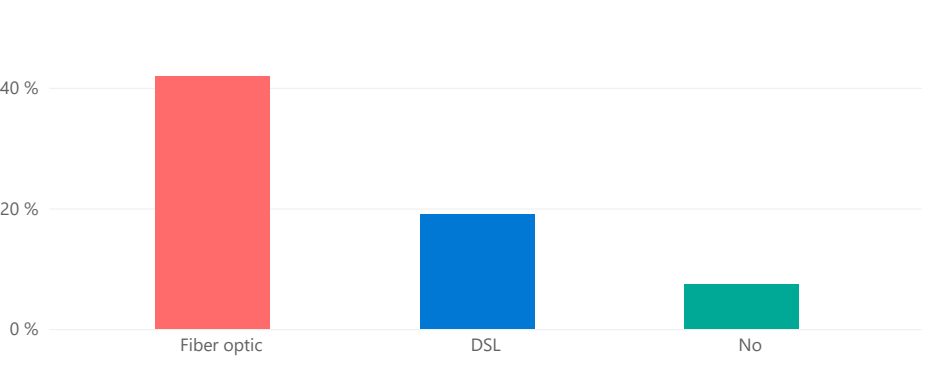
Churn Rate by Tenure Group



Churn Rate by Payment Method



Churn Rate by Internet Service



Churn Drivers

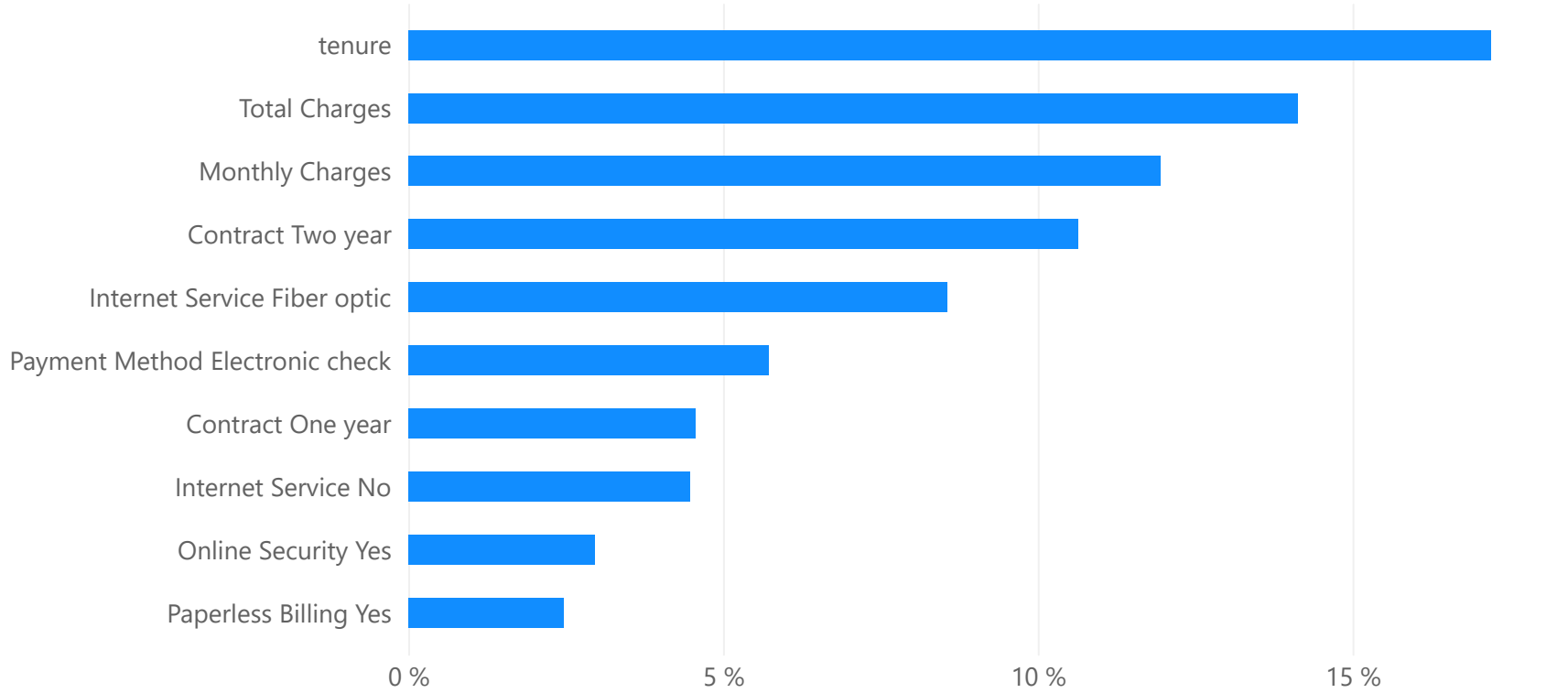
Features most associated with churn in the predictive models

Churn Overview

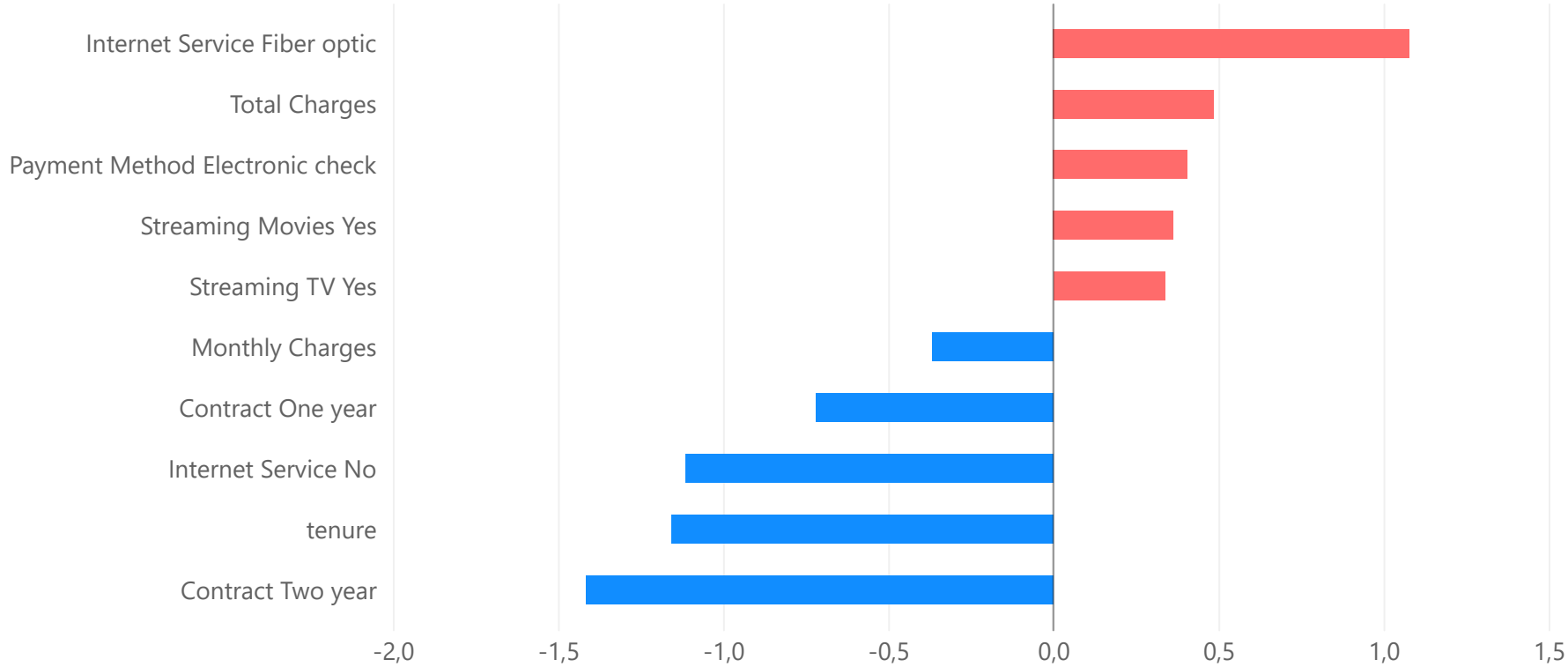
Churn Drivers

Customer Risk

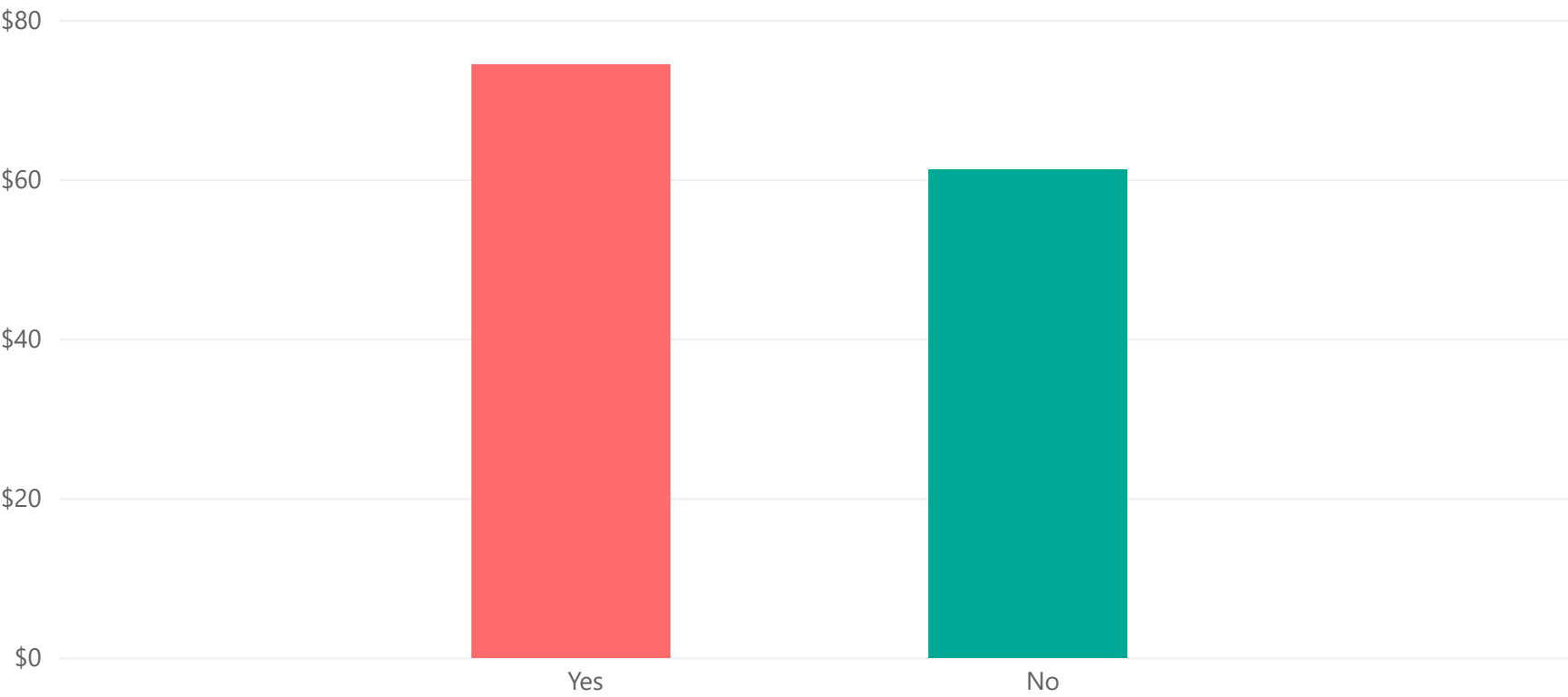
Random Forest — Feature Importance



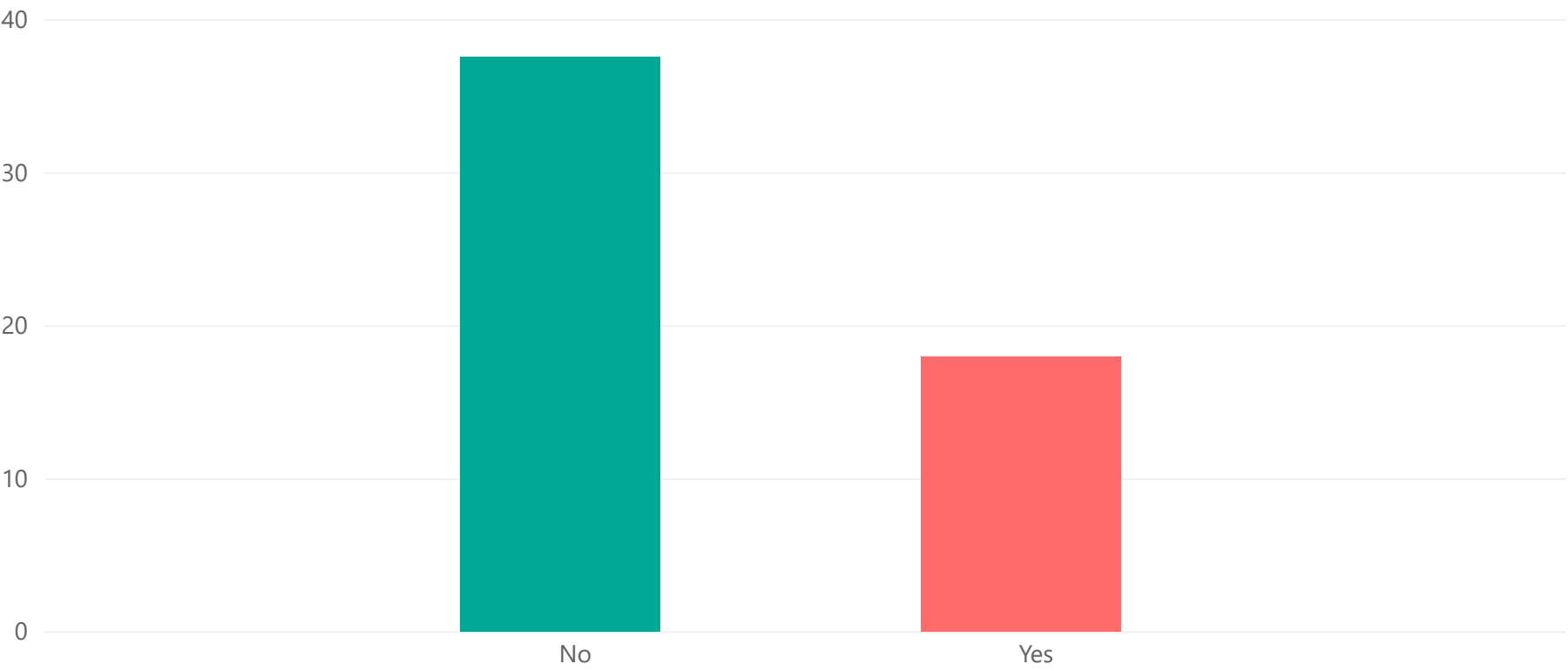
Logistic Regression — Features Associated With Churn



Average Monthly Charges — Churned vs Retained



Average Tenure — Churned vs Retained



Customer Risk

Customers classified as high risk based on predicted churn probability

High-Risk Customers

3,235

Average Churn Probability

68,6 %

High Risk Threshold

≥ 40%

Contract

☐ Month-to-month

☐ One year

☐ Two year

Internet Service

☐ DSL

☐ Fiber optic

☐ No

Payment Method

Bank transfer (automatic)

Credit card (automatic)

Electronic check

Mailed check

Tenure Group

0

72

Churn Probability

40,0 %

95,8 %

Customer	Churn Probability	Contract	Tenure (months)	Internet Service	Payment Method	Monthly Charges
7216-EWTRS	95,8 %	Month-to-month	1,00	Fiber optic	Electronic check	\$100,80
5178-LMXOP	95,7 %	Month-to-month	1,00	Fiber optic	Electronic check	\$95,10
9300-AGZNL	95,6 %	Month-to-month	1,00	Fiber optic	Electronic check	\$94,00
9497-QCMMS	95,6 %	Month-to-month	1,00	Fiber optic	Electronic check	\$93,55
5419-JPRRN	95,5 %	Month-to-month	1,00	Fiber optic	Electronic check	\$101,45
0295-PPHDO	95,2 %	Month-to-month	1,00	Fiber optic	Electronic check	\$95,45
1069-XAIEM	95,0 %	Month-to-month	1,00	Fiber optic	Electronic check	\$85,05
0107-YHINA	94,8 %	Month-to-month	1,00	Fiber optic	Electronic check	\$99,75
4910-GMJOT	94,8 %	Month-to-month	1,00	Fiber optic	Electronic check	\$94,60
8149-RSOUN	94,7 %	Month-to-month	1,00	Fiber optic	Electronic check	\$93,85
9248-OJYKK	94,7 %	Month-to-month	1,00	Fiber optic	Electronic check	\$76,45
4826-XTSOH	94,6 %	Month-to-month	1,00	Fiber optic	Electronic check	\$86,05
3068-OMWZA	94,3 %	Month-to-month	1,00	Fiber optic	Electronic check	\$88,80
4988-IQIGL	94,3 %	Month-to-month	1,00	Fiber optic	Electronic check	\$75,35
6894-LFHLY	94,3 %	Month-to-month	1,00	Fiber optic	Electronic check	\$75,75
5192-EBGOV	94,2 %	Month-to-month	1,00	Fiber optic	Electronic check	\$85,70
9282-IZGQK	94,2 %	Month-to-month	1,00	Fiber optic	Electronic check	\$85,00
8603-IJWDN	94,2 %	Month-to-month	1,00	Fiber optic	Electronic check	\$86,60
3551-HUAZH	94,1 %	Month-to-month	1,00	Fiber optic	Electronic check	\$74,40
4550-EVXNY	94,1 %	Month-to-month	1,00	Fiber optic	Electronic check	\$69,65
8779-QRDMV	94,1 %	Month-to-month	1,00	DSL	Electronic check	\$39,65
Avg	68,6 %		18,70			\$74,53